

What Happens When You Type a URL in the Browser — Simple Step-by-Step Workflow

Goal: Show in very simple terms what happens when someone types ex: `abc.com`.

Basic Architecture

Browser → Internet → DNS → Nginx Server → Backend App → Database
↓
Response goes back
↓
Browser shows page

DNS Setup (must do first)

At your domain registrar: - **Type:** A - **Name:** `abc.com` (or `@`) - **Value:** `203.45.67.89` ← Your Nginx server public IP - **TTL:** 3600

👉 This IP must be added in DNS after buying the domain. It points to your Nginx server.

Step-by-step flow

Step 1 — User types URL

- User enters `abc.com` in browser.
- Browser starts loading.

Step 2 — DNS lookup

- Browser asks: “What is the IP of `abc.com`?”
- First, it checks its local DNS cache (browser/OS). If not found, it queries the DNS system.
- DNS replies: `abc.com = 203.45.67.89`.
- This IP was added in DNS after the domain purchase. It is the Nginx server’s IP.

Step 3 — Connect to Nginx

- Browser connects to server `203.45.67.89` on port 80 (HTTP) or 443 (HTTPS).

Step 5 — Nginx checks config

- Nginx sees the request for `abc.com`.

- It matches a `server_name abc.com` block.

Step 6 — Nginx forwards to backend

- Nginx proxies the request to backend app (example: `127.0.0.1:3000`).

Step 7 — Backend processes

- Backend runs code, talks to database, builds response.

Step 8 — Database returns data

- Database gives results back to backend.

Step 9 — Response goes back

- Backend → Nginx → Browser.

Step 10 — Browser shows page

- Browser displays final content (HTML, CSS, JS).

Example Nginx config

```
server {  
    listen 80;  
    server_name abc.com;  
  
    location / {  
        proxy_pass http://127.0.0.1:3000;  
    }  
}
```